

Toward Visually Robust Quadruped Parkour via Monocular Depth Estimation

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Background

What is Quadruped Parkour?

- **Quadruped robots** have four legs and can traverse unstructured terrain
- **Parkour**: traversing discrete structured obstacles — stairs, hurdles, and gaps
- Policies are trained in **simulation via RL** and transferred to hardware
- Control runs at **50 Hz**; perception must keep up
- Key challenge: the robot must *perceive geometry* and *coordinate* that with motor commands in real time

[PLACEHOLDER]

Photo/still: real quadruped
doing parkour (Go2 or ANYmal)

Prior Work: Learning to Parkour

- **Legged Gym** [Rudin et al. 2022] — scalable sim-to-real RL framework; foundation of this work
- **Scandots** [Agarwal et al. 2023] — privileged teacher (heightmap) distilled into a depth-camera student
- **Extreme Parkour** [Cheng et al. 2023] — CNN-GRU student from depth; real-robot deployment
- **Robot Parkour Learning** [Zhuang et al. 2023] — multi-skill parkour on hardware
- **ANYmal Parkour** [Hoeller et al. 2024] — hierarchical skill selection

[PLACEHOLDER]

Figure from Extreme Parkour
(Cheng et al. 2023)

All five systems assume a **depth camera** at deployment.

Bridging Simulation to Reality: Three Strategies

1. **Domain Randomization** [Tobin et al. 2017]
Randomize textures and lighting at train time so the policy learns invariance
2. **Generative Augmentation** [Yu et al. 2024 — LucidSim]
Replace synthetic frames with diffusion-rendered photorealistic images
3. **Canonical Representation** $\leftarrow$ *this work*
Project both sim and real into a **shared depth space** via a pre-trained foundation model

[PLACEHOLDER]
Sim RGB / Real RGB $\xrightarrow{\text{DA2}}$ shared depth
(custom diagram or DA2 demo figure)

Tobin et al. 2017 | Yu et al. 2024 (LucidSim) | Yang et al. 2024 (DA2)

Research Question

Can a pre-trained monocular depth estimator replace a depth camera and enable quadruped parkour from RGB — without domain randomization?

- **Platform:** Unitree Go2 — RGB camera only, no depth sensor
- **Estimator:** Depth Anything V2 (DA2) — ViT trained on 142 M diverse images
- **Scope:** simulation study; real-world transfer is future work

[PLACEHOLDER]
Unitree Go2 photo

Mathematical Framework

Problem Formulation: POMDP

Partially Observable MDP:

$$(\mathcal{S}, \mathcal{A}, \mathcal{O}, T, R, Z)$$

- **Action** $a \in \mathbb{R}^{12}$: joint position offsets, executed via PD controller at 50 Hz
- Agent observes $o \in \mathcal{O}$, *not* full state $s \in \mathcal{S}$
- Terrain geometry is **hidden** — motivates privileged training and recurrent student architecture

Observation spaces:

Component	Dim	Content
Proprioception	53	IMU, joints, cmds
Teacher extero.	132	Scandot heightmap
Student extero.	58×87	Depth image

Partial observability $\Rightarrow$ **GRU** in student to approximate belief state

Teacher Training: Proximal Policy Optimization

PPO clipped objective:

$$\mathcal{L}^{\text{CLIP}} = \mathbb{E}_t \left[\min \left(r_t \hat{A}_t, \text{clip}(r_t, 1-\varepsilon, 1+\varepsilon) \hat{A}_t \right) \right]$$

where $r_t = \pi_\theta(a_t|o_t) / \pi_{\theta_{\text{old}}}(a_t|o_t)$

- Clipping keeps the policy update in a **trust region**
- **GAE** advantage estimate: $\hat{A}_t$ balances bias ($\lambda \rightarrow 0$) vs. variance ($\lambda \rightarrow 1$)
- Teacher has **full access** to scandot heightmap — achieves $\approx 99\%$ success across all difficulties

[PLACEHOLDER]

Teacher architecture diagram
or training reward curve
(teacher_training.pdf)

Student Distillation: DAgger

Why not behavioral cloning?

- BC on a fixed dataset $\Rightarrow$ **error compounding**
- Student mistakes shift the state distribution
— it encounters states never seen in training
 $\Rightarrow$ policy collapse

DAgger (Ross et al. 2011):

1. Roll out under **student policy**
2. Label visited states with **teacher's actions**
3. Aggregate into training set; retrain student
4. Repeat

Loss combines action imitation + yaw heading:

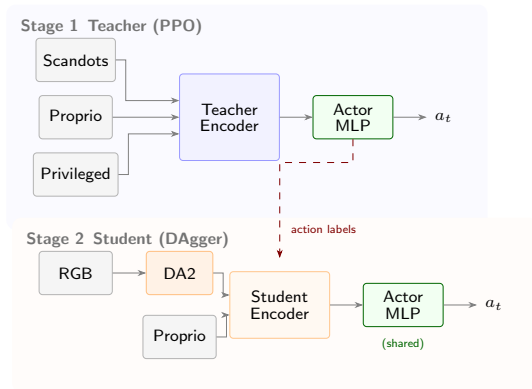
$$\mathcal{L} = \mathcal{L}_{\text{action}} + \mathcal{L}_{\text{yaw}}$$

[PLACEHOLDER]

DAgger loop diagram
(custom TikZ or Ross et al. 2011)

System Architecture

Two-Stage Teacher-Student Pipeline

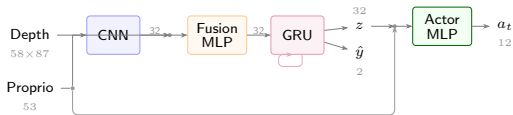


Stage 1 (PPO): Teacher trained with privileged scandot heightmap — near-perfect performance.

Stage 2 (DAgger): Student learns to mimic teacher from DA2 estimated depth images.

Student Architecture & DSQ Metric

Student visual encoder (CNN-GRU):



- 58×87 depth $\rightarrow$ CNN $\rightarrow$ 32-dim features
- GRU integrates temporally (belief state)
- Reuses teacher's frozen actor MLP

Depth Signal Quality (DSQ):

$$DSQ = \frac{1}{T} \frac{1}{N} \sum_{t,i} \rho(d_i^{GT}, d_i^{DA2})$$

where ρ is Pearson correlation.

- **Affine-invariant** — handles monocular scale-shift ambiguity
- Low DSQ $\Leftrightarrow$ degraded depth estimate $\Leftrightarrow$ policy fails
- Averaged over environments and time steps

Experiments & Results

Baselines: Upper Bounds

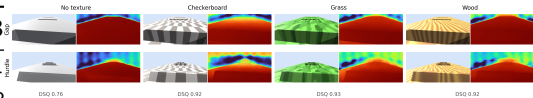
	Row 0 (easy)	Row 1	Row 2	Row 3 (hard)
Scandots teacher	98.5	100.0	99.0	98.5
GT-depth student	94.5	94.5	89.1	87.5
Zero-depth ablation	8.0	2.0	0.0	0.0

- The CNN–GRU architecture **works from depth** — GT-depth student $\approx$ teacher performance
- Zero-depth ablation confirms: policy relies on **depth, not proprioceptive heuristics** alone
- Question: can DA2 estimated depth close the remaining gap?

Success rate (%) on gap terrain, averaged over 256 episodes.

Model Scale & Terrain Texture

Configuration	R0	R1	R2	R3
DA2-small	56.3	33.6	—	—
DA2-base	68.4	83.2	26.6	0.8
DA2-base + texture	64.8	64.1	68.0	44.9
GT-depth (upper bound)	94.5	94.5	89.1	87.5



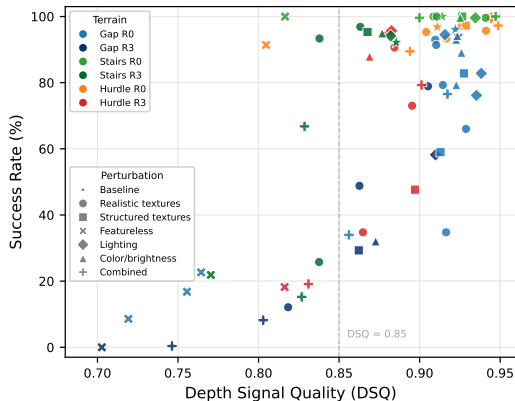
Each panel: RGB (left) and DA2 depth estimate (right). Featureless terrain produces weakest obstacle localization.

- Larger model $\Rightarrow$ better depth $\Rightarrow$ higher success
- **Untextured terrain**: gap boundary is ambiguous to DA2
- Adding checkerboard **texture** recovers most of the gap to GT-depth on harder rows

Visual Sensitivity Study

Key findings across 6 terrain configurations:

- **Stairs & hurdles:** robust to lighting and color shifts (strong edges persist across conditions)
- **Gaps:** always texture-sensitive — boundary visible mainly through surface features
- **Featureless:** catastrophic failure (96% $\rightarrow$ 17% on gap Row 0; 0% on Row 3)
- **Dim lighting** ($0.3\times$): moderate degradation only
- **Combined extreme:** 34% on gap Row 0, 0.4% on Row 3



Success rate vs. DSQ. Color = obstacle type/difficulty; shape = perturbation type.

Discussion & Limitations

What we learned:

- Monocular depth is a **viable canonical representation** for sim-to-real parkour — works in simulation
- Moderate robustness to photometric variation **without domain randomization**
- Bottleneck is **depth quality**, not student architecture
- Gap traversal the hardest case; texture is the key factor

Limitations & future work:

- **Simulation only** — no hardware validation yet
- DA2-base too heavy for onboard inference
- Frame-independent estimation — no temporal consistency
- DA2 fails on transparent surfaces, thin structures, close-range objects
- Next: real robot evaluation; joint depth + policy adaptation; temporally-aware estimators

Thank you!

Questions?